

Legal and Regulatory Issues in Life Sciences

Complete Course Notes

PART 1 FOUNDATIONS OF LIFE SCIENCES REGULATION

1.1 What the subject covers

Life sciences law governs four industries. **Pharmaceuticals** are chemically made medicines. **Biologics** are made by or from living cells, such as vaccines and insulin. **Agricultural biotechnology** covers engineered seeds and plants. **Medical devices** cover instruments, implants, diagnostics and medical software.

The science is similar in all four, but the law differs because the risk differs. **A medicine acts on one person. A modified seed acts on soil, insects, pollinators, wild plants, livestock and diets at once, and cannot be recalled once released.** That single difference explains why India clears a medicine in months and has cleared only one genetically modified crop in twenty years.

1.2 Why regulation is necessary

Four reasons. The subject of the experiment is a **living being with rights**. Science is **dual use**, so the knowledge that makes a vaccine can make a weapon. The patient and the farmer **cannot judge the science**, so regulation replaces their bargaining power. And harm may appear late or be **irreversible**, as with chlorofluorocarbon coolants and leaded petrol, both once welcomed as useful. The last reason is the basis of the **precautionary approach**, which runs through this whole subject.

1.3 Three permissions and three pillars

A drug company in India needs three separate approvals: to run the **clinical trial**, to **manufacture**, and to **market**. For a new drug all three involve the central regulator.

The regulator tests three things. **Quality** means consistent manufacturing batch after batch. **Safety** means minimum harm, shown by animal and then human studies. **Efficacy** means the drug actually works in the target patients, shown by controlled trials. **Quality, safety and efficacy are the three pillars of every drug regulator in the world.**

1.4 Growth of Indian law

Regulation began with the **Drugs Enquiry Committee** of 1930 and 1931, which produced the **Drugs and Cosmetics Act, 1940** and the **Drugs Rules, 1945**. Cosmetics were added in 1962. The **Patents Act, 1970** allowed only process patents for medicines, and that is why India became a generic drug power.

Biotechnology entered through a different door. The **Environment (Protection) Act, 1986** gave the rule making power, and the **Rules, 1989** became India's biosafety code. Note the oddity: **India regulates agricultural biotechnology through an environmental statute, not an agricultural one.**

After the World Trade Organization, India allowed product patents for medicines from 1 January 2005, passed the **Protection of Plant Varieties and Farmers' Rights Act, 2001** and the **Biological Diversity Act, 2002**, and by the Patents (Amendment) Act, 2002 allowed patents on micro organisms while excluding plants and animals.

The modern layer is mostly rules: the **Food Safety and Standards Act, 2006**, the **Medical Devices Rules, 2017**, the **New Drugs and Clinical Trials Rules, 2019**, the **Jan Vishwas Act, 2023**, the **Biological Diversity (Amendment) Act, 2023** and the **Access and Benefit Sharing Regulations, 2025**. A single consolidated drugs law has been drafted since 2022 but **has not been enacted, so the 1940 Act still governs.**

1.5 Who regulates what

For medicines and devices the central regulator is **CDSCO** under the Ministry of Health and Family Welfare, headed by the **Drugs Controller General of India**. State Licensing Authorities license ordinary drugs and inspectors enforce. The Act also creates the **Drugs Technical Advisory Board** for technical advice, the **Drugs Consultative Committee** for Centre and State coordination and the **Central Drugs Laboratory** for statutory testing. Prices are set separately by the **National Pharmaceutical Pricing Authority**.

For engineered organisms the **Ministry of Environment, Forest and Climate Change** owns the law and houses the approval committee, while the **Department of Biotechnology** houses the research committees. That department both promotes and regulates biotechnology, and this conflict of interest is what the Supreme Court criticised in 2024.

For biological resources there are three tiers: the **National Biodiversity Authority**, the **State Biodiversity Boards** and local **Biodiversity Management Committees**. Plant varieties are registered by the **PPV and FR Authority**. Research ethics is led by **ICMR**, and animal experimentation by **CCSEA**, formerly called CPCSEA.

1.6 Policy questions behind the law

Access to medicines. Patents create both the incentive to invent and the monopoly price. India did not refuse patents but built limits inside the Patents Act: section 3(d) against weak follow on patents, compulsory licensing under sections 84 to 92, the research exemption in section 107A, and price control outside the Act.

Food security. Engineered crops promise higher yield, pest resistance, drought tolerance and nutrition. Against them stand seed cost and farmer debt, corporate concentration, gene flow, and thin long term health data. A sharp point: **a consumer can choose organic or engineered food, but a farmer without capital cannot choose once rivals adopt a costlier, higher yielding technology.**

Environmental safety. Release is irreversible, so approval moves in stages from laboratory to greenhouse to small plot to multi location trial to release. A medicine acts on one species; a crop is a multi species interaction, so crop evidence takes years and never becomes conclusive.

Constitutionally, **Article 21** covers health and a clean environment, **Articles 48A and 51A(g)** cover the environment and compassion for living creatures, and **Article 47** obliges the State to raise nutrition and public health.

Vellore Citizens Welfare Forum v. Union of India, (1996) 5 SCC 647

Facts. Tanneries discharged untreated effluent into the Palar river in Tamil Nadu. A public interest petition was filed under Article 32.

Held. The petition succeeded. The Court ordered an authority under section 3(3) of the Environment (Protection) Act, 1986, imposed a fine and directed Green Benches.

Principle. **The precautionary principle and the polluter pays principle are part of the law of the land.** Lack of scientific certainty is no reason to postpone protective measures, and the **burden lies on the polluter** to show its activity is harmless. This is the bridge between environmental law and biosafety.

PART 2 RESEARCH AND INDUSTRY ETHICS

2.1 How ethics becomes law

Ethics asks what is right; law commands. Ethics changes with society and time, while law is the same for everyone at a given moment. Most of this subject is **codified ethics**. The Nuremberg Code was a statement of conscience that became a template for binding rules, and informed consent moved from an ethical norm to a statutory duty under the 2019 Clinical Trials Rules. The law converts only the settled core of ethics into obligation and leaves the rest to ethics committees.

2.2 Nuremberg and the first code

United States v. Karl Brandt and others (the Doctors' Trial), Nuremberg Military Tribunal, judgment 20 August 1947

Facts. Nazi physicians experimented on concentration camp inmates without consent, in freezing, low pressure, seawater, wound and vaccine studies. Twenty three accused were tried, including twenty doctors.

Held. Sixteen were convicted and seven acquitted. Seven were sentenced to death. The charges included conspiracy, war crimes, crimes against humanity and membership of a criminal organisation.

Principle. The section of the judgment headed *Permissible Medical Experiments* is the **Nuremberg Code**, the **first international code of research ethics**. Human experimentation is justified only if its results benefit society and it satisfies moral, ethical and legal principles. *Curiosity is never a justification.*

The ten points of the Code, in short form, are the following.

1. **Voluntary consent** of the subject is absolutely essential.
2. The study must give **fruitful results for society**, unobtainable by other means, and must not be random or unnecessary.
3. It must be based on **prior animal work** and knowledge of the disease.
4. All **unnecessary suffering and injury** must be avoided.
5. **No experiment** may be done where death or disabling injury is expected, unless the physicians are themselves the subjects.
6. The **risk** must never exceed the humanitarian importance of the problem.
7. **Proper preparation and facilities** must protect the subject even against remote risk.
8. Only **qualified persons** may conduct the work.
9. The subject may **withdraw at any time**.
10. The scientist in charge **must stop** the experiment if continuation seems likely to injure or kill.

An examination favourite: Nuremberg required **voluntary** consent, meaning consent free of pressure. Modern law requires **informed** consent, adding that the participant must be told the purpose, procedures, risks, benefits, alternatives, compensation, the right to withdraw and even any intellectual property arising from the work.

2.3 Helsinki, later abuses and Belmont

The **Declaration of Helsinki** was adopted by the World Medical Association in 1964. It is professional ethics, not State law, but legal instruments adopt it by reference. It has been revised eight times. The 1975 Tokyo revision introduced the independent ethics committee and the 2000 Edinburgh revision, with thirty two principles, is the one most textbooks describe. **The current version is the revision of 19 October 2024, and every earlier version is superseded.** The 2024 revision treats vulnerability as situational rather than as a fixed list, requires community engagement and global justice, and calls the subject a **participant**. *An answer that says Helsinki was last amended in 2000 is now wrong.*

The 1947 Code did not stop abuse. At **Willowbrook State School** children with intellectual disability were deliberately infected with hepatitis to observe the disease. At the **Jewish Chronic Disease Hospital** live cancer cells were injected into debilitated patients. In the **Tuskegee Syphilis Study**, begun in 1932 with 399 African American men, participants were misled and denied treatment even after penicillin became available, until exposure in 1972. The pattern in all three is the same: an interesting question, a population that could not refuse, and a researcher who treated the participant as a data source.

Tuskegee produced the **Belmont Report** of 1979. It has **three** principles, not four, and each carries a matching application.

1. **Respect for persons**, applied as **informed consent**.
2. **Beneficence**, applied as **assessment of risks and benefits**.
3. **Justice**, applied as **fair selection of subjects**.

The United States made Belmont binding through the **Common Rule** of 1991, requiring review by an institutional review board and documented consent. In 1982 the **Council for International Organizations of Medical Sciences**, with the World Health Organization, issued guidelines for research funded by rich countries and conducted in poor ones. Its concerns are India's concerns: consent where literacy is low, **inappropriate inducement** where payment destroys voluntariness, vulnerable populations, **therapeutic misconception** where the participant thinks the trial is treatment, and **post trial access** to a drug shown to work.

2.4 India: the ICMR guidelines

India began with the ICMR Policy Statement of February 1980, revised in 2000 and 2006, and now in force as the **National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017**, which the 2019 Rules require every researcher to follow.

The ICMR **Code of Conduct for Research Scientists in Life Sciences** states four basic principles: **autonomy**, meaning respect for persons with consent, privacy and confidentiality; **beneficence**, meaning fruitful results and doing good; **nonmaleficence**, meaning no deliberate harm; and **justice**, meaning equitable distribution of risks and benefits.

The same document lists **eleven principles of the Code**, which most class notes leave blank.

1. **Nonmaleficence**, so that no discovery harms humans, animals, plants or the environment and no one assists bioterrorism.
2. **Beneficence**, so that legitimate benefits outweigh the risks.
3. **Risk minimisation**, restricting dual use information to those who need to know.
4. **Confidentiality**, with disclosure only to legitimate persons under a contract.
5. **Ethical review** of all relevant research for ethics and safety.
6. **Transmission of ethical values** to everyone who joins the research.
7. **Voluntariness**, which here means **the researcher's right to refuse work he finds objectionable, not the participant's consent**.
8. **Compliance** with the laws applying to science.
9. **Institutional arrangements** for biosafety and biosecurity, with limited access to dangerous agents.
10. **Totality of responsibility**, spreading duty to everyone connected with the research.
11. **Research integrity**, covering reproducibility, honest data, no fabrication or plagiarism and proper authorship credit.

The Code works by a **written undertaking signed by each scientist**, not by penalty.

2.5 Ethics committees, consent and compensation

Under the **New Drugs and Clinical Trials Rules, 2019** there are two kinds of committee, one for clinical trials and one for biomedical and health research, and both must be **registered**. An unregistered committee cannot validly approve a trial. Membership mixes medical and scientific members with non scientific members, including a lay person, a woman member, a legal expert and a member independent of the institution. The committee approves the protocol in advance, supervises the trial, reviews serious adverse events and compensation, and may suspend or terminate approval.

Four numbers are worth memorising. A serious adverse event must be reported within **fourteen days**. Compensation is computed by the formula in the **Seventh Schedule**. In a trial related death the sponsor must pay within **thirty days**, besides giving **free medical management** for as long as required. The Rules also provide **post trial access** to a beneficial drug.

2.6 Leading Indian case on research ethics

Kalpana Mehta v. Union of India, (2018) 7 SCC 1 (Supreme Court, Constitution Bench, 9 May 2018)

Facts. A human papillomavirus vaccine study was run by the non governmental organisation PATH with ICMR in Khammam and Vadodara. About twenty three thousand girls aged ten to fourteen received the vaccine. Deaths and defective consent were alleged. Two public interest petitions of 2012 and 2013 challenged the regulator and ICMR, and Parliament's Standing Committee on Health had already criticised the study.

Held. The five judge Bench answered only a constitutional question. A **Parliamentary Standing Committee report may be referred to and relied upon** in proceedings under Articles 32, 136 and 226, with judicial notice under the Evidence Act, but **its correctness cannot be challenged or adjudicated**, because that would breach privilege under Articles 105, 121 and 122.

Principle. Parliamentary material is admissible as material, not as a contested issue. **The Court did not hold the vaccine trials illegal, and saying so is the classic mistake in this case.** It also shows that the practical Indian remedy for research misconduct has been public interest litigation rather than a statutory action.

2.7 Ethics of animal research

Animals are used to develop medicines and vaccines, for basic biological research and for toxicology and safety testing, including of household chemicals. Two questions must be asked first: must this project use animals at all, and must this particular test be done.

The dilemma is simple. Live animal research is at present the only route to some medicines, and treating serious illness is morally urgent, yet such research always causes suffering, and the benefit goes to humans rather than to the animals, so it must be justified as such.

Three theories answer it differently. **Contractarianism** holds that moral duties run only to those who can contract into the moral community, so on this view animal experimentation raises no ethical issue at all. **Utilitarianism**, associated with Peter Singer, counts animal suffering in the calculus, permits research where human benefit exceeds animal harm, and therefore creates a duty to reduce that harm. **The rights view**, associated with Tom Regan, is abolitionist: sentient beings cannot be treated merely as a means, and the violation is as real in promising research as in trivial research. Bernard Rollin and Arthur Caplan complete the four ethicists named in this course.

Two quotations carry the argument. Jeremy Bentham asked not *can they reason* nor *can they talk* but ****can they suffer****, which is the origin of the law's focus on sentience rather than rationality. Immanuel Kant held that man has duties only to man, because the obligating subject must be a person, though he still condemned cruelty as degrading to the human being.

The practical compromise is the **3Rs**, from Russell and Burch in 1959, and these are now legal requirements.

1. **Replacement.** Use cell culture, simulation or organ on chip systems instead of live animals wherever science allows.
2. **Reduction.** Use the statistical minimum, which is why a committee demands a power calculation.
3. **Refinement.** Better anaesthesia, less invasive procedures, humane endpoints and better housing.

Two tests are the standard hard cases. The **LD50 test** finds the dose that kills half the test population, and is used not only for medicines but for cleaners and pesticides, the resulting data being what makes poisoning treatment possible. The **Draize test** applies a substance to the eye or skin of an animal, usually an albino rabbit, to test irritancy. The examination question is whether banning them is justified even if human deaths from toxic exposure rise. Answer it as a conflict of interests resolved through the 3Rs and validated alternatives, not by choosing a side.

2.8 Law on animal research

In the United States the **Animal Welfare Act** of 1966, amended in 1985, requires societal benefit to be balanced against the imposition on animals, minimum pain and distress, species appropriate housing, veterinary involvement, qualified investigators and oversight by an **Institutional Animal Care and Use Committee**, which applies three tests: a clear rationale for animal use, justification of the number used, and a

risk and benefit analysis. A sharp critique: the official count of about 1.1 million animals used in 2009 **excludes rats and mice**, which are most laboratory animals, because they fall outside the Act.

In India the statute is the **Prevention of Cruelty to Animals Act, 1960**. **Section 15** constitutes the committee now called **CCSEA**, with powers in section 17, and its mandate is to ensure that animals suffer no unnecessary pain **before, during or after** experiments. The working rules are the **Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998**, amended in 2001 and January 2006.

Compliance has four parts. An establishment doing biomedical research, or breeding or trading laboratory animals, must **register with CCSEA**, constitute an **Institutional Animal Ethics Committee** including a CCSEA nominee, have its animal house inspected, and obtain approval for every project. **Approval by an improperly constituted committee is invalid.** Animal welfare moved to the Department of Animal Husbandry and Dairying in April 2019, so the name CPCSEA and the old ministry are both out of date.

Animal Welfare Board of India v. A. Nagaraja, (2014) 7 SCC 547 (Supreme Court, 7 May 2014)

Facts. Jallikattu in Tamil Nadu and bullock cart races in Maharashtra were challenged, along with the Tamil Nadu Regulation of Jallikattu Act, 2009, after a notification of 11 July 2011 barred the use of bulls as performing animals.

Held. The events as then conducted violated sections 3 and 11 of the 1960 Act. The notification was upheld and the State Act was held to offend the central Act.

Principle. The 1960 Act must be read with **Articles 51A(g), 48A and 21**. The Court adopted the **five internationally recognised freedoms** and held that **animals have a right to live with dignity**.

Animal Welfare Board of India v. Union of India, 2023 SCC OnLine SC 661 (Supreme Court, Constitution Bench, 18 May 2023)

Facts. Tamil Nadu, Maharashtra and Karnataka amended the 1960 Act to permit Jallikattu, bullock cart races and Kambala with safeguards. The amendments were challenged.

Held. The amendments were **upheld**. They substantially reduce pain, fall within **Entry 17 of the Concurrent List**, and cure the defects identified earlier. The cultural rights question under Article 29(1) was left open.

Principle. ***Nagaraja* was confined and distinguished, not overruled.** Read the two together: **Indian law protects animal welfare rather than conferring animal rights**, and a legislature may lawfully restore a practice by curing the defects a court has identified.

2.9 Bioethics

Bioethics is ethics applied to the life sciences. It covers research ethics for humans and animals, clinical ethics such as consent and resource allocation, public health ethics such as vaccination mandates, and the ethics of new technology such as genome editing, biobanking and artificial intelligence in diagnosis. **Dual use research of concern** is its sharpest edge. Bioethics enters patent law through **section 3(b)** of the Patents Act, which excludes inventions contrary to public order or morality or seriously prejudicial to life, health or the environment. **Do not confuse section 3(b), the morality clause, with section 3(d), which is about new forms of known substances.**

PART 3 AGRICULTURAL BIOTECHNOLOGY AND SEED REGULATION

3.1 Basic definitions

A genetically modified crop carries a gene or constructed DNA sequence introduced by modern biotechnology rather than by breeding, often from an unrelated species. **Bt crops** carry genes from *Bacillus thuringiensis* so that the plant makes a Cry protein toxic to particular insect larvae; in Bt cotton the target is the bollworm. **Herbicide tolerant crops** survive a broad spectrum herbicide so the field can be sprayed without killing the crop.

Genome edited crops are different in kind. Techniques such as CRISPR alter the plant's own genes without inserting foreign DNA in the categories called SDN 1 and SDN 2, and this distinction now drives Indian policy. A **living modified organism**, the term used by the Cartagena Protocol, is narrower than a genetically modified organism, because it must be **alive and able to transfer or replicate genetic material**. Processed food made from such an organism is not itself one.

3.2 The case for and against

The claimed traits are pest resistance, higher yield, drought tolerance and nutritional fortification, on landholdings averaging about one hectare. Four objections answer them.

1. **Economic.** Engineered seed costs far more, farmers borrow to buy it, and if the crop fails for unrelated reasons the debt remains. State this as an asserted link to farmer distress, not proven causation, because the evidence is divided.
2. **Market structure.** A few companies control most commercial seed, because research and the regulatory dossier are costly. That is a competition problem needing price control and licensing, not a biosafety answer.
3. **Environmental.** Loss of non target insects and pollinators in monocultures, gene flow to conventional crops and wild relatives, and pest resistance that erodes the benefit over time.
4. **Health.** No study conclusively shows harm from approved engineered food, though some animal studies report effects. **Absence of proof of harm is neither proof of safety nor a reason to prohibit**, and that gap is where the precautionary approach operates.

3.3 Indian framework and the six committees

The parent statute is the **Environment (Protection) Act, 1986** and the operative code is the **Rules, 1989**, which apply to engineered organisms, micro organisms and cells and to any substance, product or foodstuff of which they form part, so the Rules follow the organism into the product. **India regulates by the technique used, like the European Union, not by the characteristics of the final product, like the United States.** That one design choice explains most of India's difficulties.

The Rules create **six competent authorities**, and the examinable point is the function of each. *Many class notes wrongly say four.*

1. **RDAC**, the Recombinant DNA Advisory Committee, in the Department of Biotechnology. Purely **advisory**; it reviews developments and recommends safety regulations.
2. **IBSC**, the Institutional Biosafety Committee. The **institutional regulator**, formed by every institution handling such organisms, with the head of the institution, the scientists, a medical expert and a nominee of the Department of Biotechnology. It fixes risk class and containment and prepares the emergency plan.
3. **RCGM**, the Review Committee on Genetic Manipulation. It **monitors ongoing research** and clears small scale and **confined field trials** that generate the biosafety data.
4. **GEAC**, the Genetic Engineering Appraisal Committee, under the environment ministry. The **approval** body for large scale use and for **release into the environment**. Nothing containing such an organism may be produced, sold, imported or used without its permission.
5. **SBCC**, the State Biotechnology Coordination Committee. **Monitoring** at State level.
6. **DLC**, the District Level Committee. **Monitoring** at district and installation level.

Memory device: advisory, institutional, research monitoring, approval, and two monitoring tiers. The approval pathway runs in the same order.

1. The **institutional committee** fixes the risk group and containment. A soil bacterium can be handled anywhere, while a tuberculosis organism or a virus of the COVID class needs high containment.
2. The **research committee** clears contained work and confined field trials and monitors them.
3. Trials **expand in area in stages**, because an open environment cannot be modelled in a greenhouse.
4. The **appraisal committee** decides on environmental release or commercial cultivation.

5. The **State and district committees** monitor afterwards.
6. Engineered **food** needs a further approval from the **Food Safety and Standards Authority of India**, and imports also engage foreign trade policy and plant quarantine rules.

3.4 Seed laws and the genome editing exemption

The **Seeds Act, 1966** and the **Seeds (Control) Order, 1983** govern seed quality and trade, not biosafety, and do not define a genetically modified organism, which is why a separate regime was needed. The **Cotton Seed Price (Control) Order, 2015**, under the Essential Commodities Act, 1955, fixes the maximum seed price and the trait value a technology provider may charge. That Order is the direct cause of the Monsanto litigation in Part 5.

In 2022 the environment ministry **exempted genome edited plants in the SDN 1 and SDN 2 categories** from the strictest provisions of the Rules, 1989, because they contain no foreign DNA, while SDN 3 remains fully regulated. The institutional committee must certify absence of foreign DNA. Two genome edited rice lines were announced in May 2025. The examinable criticism is twofold: India now uses a **product based approach for genome editing while keeping a process based approach for transgenics**, which is inconsistent, and it did so by **executive action rather than by amending the Rules**.

3.5 The three Indian crops

Bt cotton was approved in 2002 and **remains the only genetically modified crop approved for commercial cultivation in India**. Many varieties carry the trait, so one crop does not mean one product.

Bt brinjal was recommended for release by the appraisal committee on **14 October 2009**. After a scientific review and public hearings in seven cities, the environment minister imposed an **indefinite moratorium on 9 February 2010** on a precautionary basis, to last until independent studies establish safety to the satisfaction of the public and the scientific community. He clarified a week later that this was a moratorium and not a ban. The drug regulator and ICMR had asked for more food safety testing, and major brinjal growing States had asked for deferment, which shows the federal dimension. It was then that the committee was renamed from *Approval* to *Appraisal*, signalling that the technical body appraises while the government decides. The moratorium has never been lifted.

Genetically modified mustard, the hybrid DMH 11, was developed at Delhi University using the barnase and barstar system with a herbicide tolerance gene. The appraisal committee recommended conditional release on **18 October 2022** and the ministry conveyed the decision on **25 October 2022**, which would have made it India's first transgenic food crop. Litigation followed, including the long pending petition of Aruna Rodrigues, in which the Supreme Court had appointed a **Technical Expert Committee** in 2012 whose 2013 report criticised the regulatory system, found inherent conflicts of interest and recommended a ten year moratorium on engineered food crops.

Gene Campaign v. Union of India, 2024 INSC 545 (Supreme Court, 23 July 2024)

Facts. Public interest petitions challenged the approval for environmental release of genetically modified mustard DMH 11 and, more broadly, the adequacy of India's biosafety regulation.

Held. A **split verdict**. Justice Nagarathna quashed the approval for breach of the statutory procedure, the absence of an ICMR representative at a crucial meeting and the sidelining of the Technical Expert Committee. Justice Karol upheld it, holding the appraisal committee an expert body whose conditional approval showed no non application of mind. The matter went to the Chief Justice for a larger Bench, so **the approval is neither quashed nor implemented and cultivation remains on hold**.

Principle. Both judges agreed on five directions, and these form the operative part of the case.

1. **Judicial review** of decisions on genetically modified organisms **is permissible**, so the regulator's scientific judgement is not immune.
2. The Union **must frame a national policy** on genetically modified crops covering research, cultivation, trade and commerce.

3. That policy must follow a **national consultation within four months** with all stakeholders including States and farmers.
4. Experts' credentials must be **verified** and **conflicts of interest mitigated**, with rules framed for the purpose.
5. The **Food Safety and Standards Act** must be enforced on imports of engineered food.

Do not write that the Court banned genetically modified mustard, and do not write that it approved it. It split and referred the matter on.

3.6 United States and European Union compared

The United States has no dedicated biotechnology statute. Under the **Coordinated Framework of 1986** three agencies use existing powers: the agriculture department treats the plant as a possible plant pest, the environment agency treats the Bt trait as a pesticide, and the food and drug agency examines food and feed safety. The result is faster approval and a long list of approved products. The **StarLink corn incident of 2000**, where a feed only variety entered the human food chain, is the standard example of a segregation failure.

The European Union regulates by process, with a precautionary principle in its founding Treaty, through four instruments.

1. **Directive 2001/18**, requiring authorisation, environmental risk assessment, monitoring and time limited consents for deliberate release.
2. **Regulation 1829/2003**, giving a single authorisation for engineered food and feed after a scientific opinion.
3. **Regulation 1830/2003**, imposing **labelling and traceability** through the supply chain.
4. **Directive 2015/412**, allowing a Member State to **opt out of cultivation** of a product authorised at Union level on grounds other than risk.

Confédération paysanne v. Premier ministre, Court of Justice of the European Union, Grand Chamber, 25 July 2018

Facts. French farmers' and environmental groups asked whether plants produced by newer mutagenesis techniques, including gene editing, fall within the European Union's regime for genetically modified organisms. The Conseil d'Etat referred the question.

Held. Organisms obtained by **mutagenesis are genetically modified organisms** under Directive 2001/18. The exemption in Annex I B covers only mutagenesis techniques **conventionally used with a long safety record**, so newer directed techniques are fully regulated. Advocate General Bobek had advised the opposite.

Principle. Regulation follows the **technique**, not the product. **Set this against India's 2022 exemption of SDN 1 and SDN 2: two jurisdictions reached opposite answers on the same scientific question.**

The European Union is now legislating a more permissive framework, so check the current position before treating 2018 as final.

3.7 The World Trade Organization case

European Communities, Measures Affecting the Approval and Marketing of Biotech Products, WT/DS291/R and connected reports, adopted 21 November 2006

Facts. The United States, Canada and Argentina complained that the European Communities had stopped approving biotech products, delayed individual applications and permitted national bans on already approved products. The panel was established on 29 August 2003 and the reports were circulated on 29 September 2006. There was no appeal.

Held. Three findings. A **general de facto moratorium** existed from **June 1999 to August 2003**, which the European Communities had denied; it was not itself a sanitary measure but it caused failure to complete approval procedures **without undue delay**, contrary to **Annex C(1)(a) and Article 8 of the SPS Agreement**. There was undue delay in **twenty four of the twenty seven** products identified. Nine

national safeguard measures breached **Articles 5.1 and 2.2** because they rested on no risk assessment and could not be saved as provisional measures under Article 5.7.

Principle. **A Member may regulate genetically modified products strictly, but it may not simply refuse to decide.** The panel did not rule that such products are safe, and did not rule on the European approval legislation itself, which was never challenged. It ruled on **procedural delay** and on the **evidentiary basis of national bans**. India is exposed in the same way: an indefinite moratorium with no formal, reasoned, science based refusal is legally weaker than an express refusal.

3.8 Cartagena Protocol on Biosafety

The **Cartagena Protocol** supplements the Convention on Biological Diversity and was negotiated under Article 19(3) of the Convention. It was **adopted on 29 January 2000** and **entered into force on 11 September 2003**, and **India is a party since 2003**. Its objective is the safe transfer, handling and use of living modified organisms.

It applies to the transboundary movement, transit, handling and use of such organisms that may adversely affect the conservation and sustainable use of biological diversity, taking into account risks to human health. Three categories are covered and two are exempt.

1. Organisms for **direct use as food or feed or for processing**, such as imported soya for crushing.
2. Organisms for **contained use**, such as a strain imported for laboratory research.
3. Organisms for **intentional introduction into the environment**, such as seed for cultivation.
4. **Exempt**: pharmaceuticals for humans covered by other international agreements.
5. **Exempt**: processed products that no longer contain a living modified organism.

The spine of the Protocol is the **Advance Informed Agreement** procedure, which applies before the first intentional transboundary movement of an organism meant for release into the environment. It has four steps.

1. **Notification by the exporter**, being a detailed written description given well before the first shipment.
2. **Acknowledgement of receipt** by the importer.
3. A **decision** to approve, prohibit, impose conditions or seek more information.
4. **Review of the decision** if circumstances change.

The procedure does not apply in the same way to food, feed and processing shipments or to contained use, and that limitation is a common examination trap. The Protocol also requires **risk assessment and risk management**, documentation for shipments, and emergency notification for illegal or unintentional movements.

Information flows through the **Biosafety Clearing House** under Article 20(1), which exchanges scientific, technical, environmental and legal information and assists developing country parties, least developed countries, small island States and centres of origin and genetic diversity. It keeps records in the six official languages and maintains a **Unique Identifiers Registry** of every registered organism, a **Gene Registry** of inserts and modifications, and an **Organism Registry** of parental, recipient and donor organisms. The last one links biosafety to benefit sharing, because benefits cannot be shared unless the origin of a resource can be traced.

The Protocol expressly **reaffirms Principle 15 of the Rio Declaration** on precaution, and is supported by capacity building, socio economic considerations, a compliance procedure and the liability regime later developed in the Nagoya Kuala Lumpur Supplementary Protocol of 2010. Developing country parties are eligible for funding from the **Global Environment Facility**. One point often asked: the Protocol governs movement **between** countries, while movement **within** a country is governed by domestic law, institutional clearance and a **Material Transfer Agreement**, which allocates risk privately as the Protocol does internationally.

3.9 Terminator technology

Genetic Use Restriction Technology, popularly the terminator gene, restricts use of a plant or its progeny, in its strong form by making harvested seed sterile so that the farmer must buy fresh seed each season. Indian law blocks it through two provisions of the Protection of Plant Varieties and Farmers' Rights Act, 2001.

1. **Section 18(1)(c)**. Every application for registration must carry an **affidavit that the variety contains no gene or gene sequence involving terminator technology**.
2. **Section 29(3)**. No variety involving technology injurious to the life or health of humans, animals or plants may be registered, and the Explanation says *any technology includes genetic use restriction technology and terminator technology*.

Note the precise legal point. **The Act denies registration; it is not itself a general prohibition on developing or importing such a variety**. A prohibition on release would come from the biosafety Rules. The law takes this shape because terminator technology attacks the right Indian law guards most closely, the farmer's right to save and resow seed, and a technology that makes a legal right physically impossible is treated as inconsistent with the statute.

PART 4 PHARMACEUTICAL AND BIOPHARMACEUTICAL REGULATION

4.1 Phases of a clinical trial

1. **Preclinical**. Laboratory and animal study of pharmacology, toxicology and preliminary safety.
2. **Phase I**. First in human, a small number of healthy volunteers, testing safety, tolerability and dose. Patients are used instead for cytotoxic drugs.
3. **Phase II**. A few hundred patients, for preliminary efficacy and dose finding.
4. **Phase III**. Large, usually randomised and controlled, often multi centre. This supplies the evidence for marketing approval.
5. **Phase IV**. After marketing, meaning **pharmacovigilance** for rare adverse events, long term effects and excluded populations.

A **bioequivalence study** is not a phase. It is the comparative study a generic applicant uses to show that its product behaves in the body like the original.

4.2 Drugs and Cosmetics Act, 1940

The Act regulates the import, manufacture, distribution and sale of drugs and cosmetics so that what is sold in India is safe, effective and of standard quality. Section 3(b) defines **drug** widely, covering medicines for humans and animals, substances used in diagnosis, treatment, mitigation or prevention of disease, substances affecting the structure or function of the body, components of drugs, and devices notified as drugs. Chapter III covers import, Chapter IV covers manufacture, sale, distribution and most offences, and Chapter IVA covers Ayurvedic, Siddha and Unani drugs.

The four categories of defective drug drive the penalties and are the most examinable definitions in the statute.

1. **Not of standard quality**, section 16. The drug fails the prescribed standard of strength, quality or purity. A manufacturing failure and the commonest category.
2. **Misbranded**, section 17. Coloured or coated to hide damage, made to look better than it is, not labelled as prescribed, or carrying a false or misleading claim.
3. **Adulterated**, section 17A. Contains filthy or decomposed matter, made or stored in insanitary conditions, in a poisonous container, or contains a harmful substance.
4. **Spurious**, section 17B. Deceives as to identity or source: sold under another drug's name, imitating or substituting another drug, naming a fictitious manufacturer, or falsely purporting to come from a maker.

Not of standard quality is a manufacturing failure; spurious is a fraud. The whole debate about decriminalisation is whether the first deserves leniency when the consequence for the patient can be identical.

4.3 Offences and the Jan Vishwas Act, 2023

Section 27 grades punishment by seriousness, the heaviest being for adulterated or spurious drugs likely to cause death or grievous hurt, with **clause (d)** as the residuary clause covering misbranding and substandard quality. Section 27A does the same for cosmetics. **Section 32B** allows **compounding** of specified offences, not being offences punishable with imprisonment only, on payment of a sum not exceeding the maximum fine, before or after prosecution begins, with subsequent offences expressly not compoundable.

The **Jan Vishwas (Amendment of Provisions) Act, 2023**, assented on **11 August 2023**, amended 183 provisions across 42 Central Acts to decriminalise minor offences. It made three changes to the Drugs and Cosmetics Act, in **sections 29, 30 and 32B**. The important one is that **section 27(d) and section 27A(ii) were added to the list of compoundable offences**, so misbranding and not of standard quality can now be settled by payment instead of prosecution. The amendments took effect from **31 December 2024**, and minimum fines rise by ten per cent every three years, the first escalation falling due at the end of 2027.

Argue both sides. Critics say a manufacturer whose substandard medicine injures a patient can now avoid punitive accountability, that substandard quality is no technical lapse when a drug that fails assay fails the patient completely, and that there is a drafting tension between compounding under section 32B and imprisonment under section 27(d). The government says fear of imprisonment for minor lapses deters legitimate business, that adulterated and spurious drugs causing death remain fully criminal, and that monetary penalties are faster and more certain than long prosecutions.

4.4 The draft consolidated law

A **Drugs, Medical Devices and Cosmetics Bill** has been drafted repeatedly, in 2022, 2023, a 2025 draft presented after children died from a contaminated cough syrup, and a 2026 draft under inter ministerial consultation. **It is not law. The 1940 Act with the 2017 Device Rules and the 2019 Trials Rules remains in force**, and getting this status right is the likeliest trap in a question on the subject.

Its main features are that **medical devices become a separate category** with their own classification, licensing and clinical investigation provisions; inspectors become **Drugs Control Officers** able to issue an **Improvement Notice** before prosecution; **online pharmacies and AYUSH integration** get statutory recognition; and the central regulator gains **statutory powers over quality testing and market surveillance**. Eleven industry associations have instead asked for a **standalone Medical Devices Act** and an independent regulator, arguing that a pharmaceutical centred law criminalises technical and labelling lapses in an engineering industry even where no patient is harmed. A good answer notes both that reform is overdue, because the 1940 Act predates biologics, devices and software, and that consolidation reproduces the very conflation the 2017 Rules were meant to escape.

4.5 Medical Devices Rules, 2017

The Rules were made under the 1940 Act and took effect from 1 January 2018. By notification of **11 February 2020** every medical device was notified as a drug, bringing the whole sector within the net. Devices are classified by risk.

1. **Class A**, low risk, such as surgical dressings. Licensed by the State authority.
2. **Class B**, low moderate risk, such as hypodermic needles. Licensed by the State authority.
3. **Class C**, moderate high risk, such as ventilators and bone plates. Licensed centrally.
4. **Class D**, high risk, such as heart valves and coronary stents. Licensed centrally, as are all imports.

Three features matter. Licences are **perpetual** unless cancelled or surrendered, subject to a **retention fee every five years**, which replaced periodic renewal and is one of the real liberalisations in Indian regulation. Quality systems are assessed by accredited **Notified Bodies** for the lower classes, in practice against ISO 13485. And labelling is device specific and includes a **unique device identifier** for traceability, with **Rule 96** giving the Device Rules **primacy** over the old Drugs Rules where they conflict.

Recent amendments recognise a registration and self declaration route for low risk Class A devices, align device labelling with the Legal Metrology Rules of October 2025, compress scrutiny timelines in a 2026 draft, and add draft guidance on **software as a medical device**. That last point matters: **an artificial intelligence diagnostic tool sold in India is regulated as a medical device by its risk class, not as ordinary software.**

4.6 New Drugs and Clinical Trials Rules, 2019

Notified as G.S.R. 227(E) on **19 March 2019**, these Rules superseded Part XA and Schedule Y of the old Drugs Rules and now govern new drugs, investigational new drugs, clinical trials, bioequivalence studies and ethics committees.

A **new drug** is any of three things.

1. A drug **not used in India to any significant extent** and not approved as safe and efficacious by the central authority for its claims.
2. An **approved drug proposed to be marketed with a new claim**, indication, route, dosage or dosage form.
3. A **fixed dose combination** of separately approved drugs combined for the first time or in a new ratio.

So a new route of administration, or a pentavalent vaccine combining five known vaccines in one injection, is a new drug even though every component was already approved.

A drug remains a new drug for **four years** from first approval or inclusion in the Indian Pharmacopoeia, whichever is earlier. Two results follow: the central authority rather than the State controls it, and a later applicant cannot simply rely on the originator's dossier. **That four year window is India's functional substitute for data exclusivity, which India has never enacted.** An **investigational new drug** is a new chemical entity or product with a therapeutic indication never before tested on human beings.

Timelines are **thirty working days** for a drug discovered, developed and to be manufactured in India, with **deemed approval** if the authority says nothing, and **ninety working days** otherwise. A **local clinical trial may be waived** for a drug already approved and marketed in specified countries. A 2024 amendment brought **clinical research organisations** under mandatory registration, closing the gap that arose because trials are outsourced to them.

Argue the waiver both ways. Against it, differences in metabolism, body weight, diet, comorbidity and disease strain mean foreign data may not predict Indian response. For it, insisting on complete local trials in an emergency such as COVID costs lives. The law's answer is a conditional waiver backed by Phase IV surveillance.

4.7 Generics, biologics and biosimilars

A **generic** is comparable to the innovator product in dosage form, strength, route, quality, performance and intended use. Its application is **abbreviated** because animal and human safety and efficacy studies need not be repeated. What must be proved is **bioequivalence**: give both products, measure blood concentration over time, and show that the rate and extent of absorption fall within accepted limits, from which therapeutic equivalence is inferred.

A sharp Indian point: for one molecule the market carries the innovator at the top price, established Indian brands in the middle and lesser known generics at the bottom, and the middle tier charges its premium on the strength of its **trade mark**, not any patent. **In India trade mark law, not patent law, is often the real source of pricing power in the generic market,** which is why prescribing by generic name and price control are the policy levers.

Biologics are made from living sources, human, animal or microbial, and include vaccines, blood products, allergenic extracts, tissues for transplantation, gene therapies and cell therapies. A large molecule made by a living cell line depends on that line, the culture conditions and the purification process, so two manufacturers cannot make an identical molecule. Hence **similar**, not **same**, and hence a separate pathway.

A **biosimilar** must satisfy two limbs. **Highly similar** is shown by extensive characterisation of structure and function, comparing purity, chemical identity and bioactivity, with **minor differences in clinically inactive components** such as the buffer being acceptable. **No clinically meaningful differences** is shown in safety, purity and potency through human pharmacokinetic and pharmacodynamic studies, an assessment of **clinical immunogenicity** and, if needed, further clinical studies. Immunogenicity is the distinctive risk, because a large foreign protein can provoke an immune response.

An **interchangeable** product, a concept from United States law, is a biosimilar that additionally shows it will give the same clinical result in any given patient and that switching back and forth has been evaluated, which allows substitution at the counter. **Every interchangeable product is a biosimilar, but not every biosimilar is interchangeable.** And the difference from a generic is exact: a generic has the **same** active ingredient and proves **bioequivalence**, while a biosimilar is **highly similar** and must additionally prove no clinically meaningful difference. Because development is costlier and slower, biosimilar entry cuts price less than generic entry, which is why India leads in chemical generics but not in biologics.

India regulates biologics largely through guidance: the Recombinant DNA Safety Guidelines of 1990, the 1999 guidelines on data for recombinant products, the CDSCO industry guidance of 2008, the Institutional Biosafety Committee handbook of 2011 and the **Guidelines on Similar Biologics of 2012**. Two regulators apply at once, the drug regulator under the 1940 Act and the recombinant DNA authorities under the Rules, 1989. A strong criticism follows: **India regulates its most complex medicines mainly through non statutory guidelines.**

One advanced point. A chemical drug may carry one or a few patents, while a biologic is covered by a thicket over the sequence, the cell line, the expression system, the formulation, the process, the method of treatment and the device. A broad early claim to a class of molecules is loosely called a **genus** claim and a later narrow claim a **species** claim, so a biosimilar developer who designs around one may fall within the other, and the dispute becomes one of **claim construction**. Infringement analysis therefore has two steps: construe the claims, then ask whether the product falls within them as construed.

4.8 Harmonisation and comparison

The **Common Technical Document** is the accepted dossier format, with five modules: **regional, summaries, quality, nonclinical** and **clinical**. It comes from the **International Council for Harmonisation**, whose E6 guideline on good clinical practice underlies Indian practice. Harmonisation reduces but does not remove mismatch, because Module 1 stays national and acceptance of foreign data remains a sovereign choice. Finished product specifications also still differ across countries in assay limits and other tests, so a product can be lawful in one and unlawful in another with no difference in safety, which is a pure regulatory barrier to trade.

The United States built its system through five statutes.

1. **Pure Food and Drugs Act, 1906**, answering widespread adulteration. Note that food and drugs were regulated together from the start, unlike India.
2. **Food, Drug and Cosmetic Act, 1938**, after the sulfanilamide disaster, adding premarket safety review.
3. **Kefauver Harris Amendment, 1962**, after thalidomide, adding proof of **efficacy** and consent of research subjects. This is when the three pillar model became law.
4. **Hatch Waxman Act, 1984**, creating the abbreviated generic application, the regulatory testing exemption, patent term restoration and **patent linkage**.
5. **Biologics Price Competition and Innovation Act, 2009**, creating the biosimilar and interchangeable routes.

The European Union instead runs a network, in which the European Medicines Agency supplies the science while the Commission grants the authorisation, through four routes: the **centralised** procedure giving one authorisation across the Union and mandatory for biotechnology products and orphan medicines; the **decentralised** procedure with one reference Member State; **mutual recognition** of an existing national authorisation; and a purely **national** route.

The contrast is instructive. A single regulator trades speed for distance from local conditions, a network preserves autonomy at the cost of complexity, and India's Centre and State split suffers from neither but from **uneven enforcement capacity**, which is what the draft Bill tries to fix.

4.9 Patent linkage, compulsory licensing and data exclusivity

Patent linkage ties the marketing approval of a generic to the patent status of the original drug, so that in its strong form the drug regulator cannot approve a generic before the patent expires without the patentee's consent.

Bayer Corporation v. Union of India, 2010 (43) PTC 12 (Del) (Delhi High Court, Division Bench, 9 February 2010)

Facts. Bayer held an Indian patent on **sorafenib tosylate**, sold as Nexavar for advanced kidney cancer. When Cipla applied for marketing approval of a generic version in 2008, Bayer asked the Drugs Controller General to refuse, arguing that section 2 of the Drugs and Cosmetics Act read with sections 48 and 156 of the Patents Act obliged the regulator to protect its patent, and that the generic would be a spurious drug.

Held. The writ petition was dismissed by a Single Judge on 18 August 2009, the appeal was dismissed by the Division Bench, and the Supreme Court dismissed the further appeal in December 2010. The two statutes have **different objects and operate in different spheres**: the Drugs Act keeps unsafe, adulterated, misbranded and spurious drugs off the market, while the Patents Act lets a patentee restrain third parties. The regulator is **not equipped to decide patent disputes**, and accepting Bayer's reading would block generic entry by requiring the regulator to presume every patent valid.

Principle. **Patent linkage is not part of Indian law and cannot be read into the Drugs and Cosmetics Act.** A patentee is not without remedy; it must sue for infringement and prove its case.

Do not confuse that case with the **compulsory licence** on the same drug, which is a separate line of proceedings.

Bayer Corporation v. Union of India, 2014 (60) PTC 277 (Bom) (Bombay High Court, 15 July 2014)

Facts. Natco sought a compulsory licence under **section 84** for sorafenib tosylate. Bayer's price was about Rs 2,84,428 for a month's therapy against Natco's offer of about Rs 8,880. The Controller granted the licence on 9 March 2012 at a six per cent royalty and the appellate board upheld it on 4 March 2013, raising the royalty to seven per cent.

Held. Bayer's writ petition was dismissed, and the Supreme Court declined to interfere on 12 December 2014. **All three grounds under section 84(1) were satisfied**: the reasonable requirements of the public were not met, the drug was not available at a reasonably affordable price, and the patent was not worked in India. A patient assistance programme was held to be charity, not an answer.

Principle. **India's first compulsory licence.** Whether a patent is **worked in the territory of India** is a fact dependent question and does not always require local manufacture, but on these facts Bayer had not worked the patent adequately.

Data exclusivity is the live issue. Article 39.3 of the TRIPS Agreement requires protection of undisclosed test data submitted for marketing approval against **unfair commercial use**. India reads that as protection against unfair use and disclosure, not as a fixed exclusivity term, and relies instead on the four year new drug window. India resists exclusivity because it works **independently of patents**: a drug that is unpatented in India, or refused a patent under section 3(d), could still be shielded, since a generic applicant could neither use the original data nor afford to repeat the trials. That would undo much of what section 3(d) and compulsory licensing achieve.

Finally, prices are capped for scheduled formulations under the **Drugs (Prices Control) Order, 2013**, with power to fix the price of any drug in extraordinary circumstances in the public interest, used for cardiac stents and knee implants. Safety after marketing is watched through the **Pharmacovigilance Programme of India**, because no trial of a few thousand patients can detect the rare and the late.

PART 5 INTELLECTUAL PROPERTY RIGHTS AND BIOTECHNOLOGY

5.1 TRIPS and the unfinished review

The TRIPS Agreement of 1994 binds every member of the World Trade Organization and is enforceable through dispute settlement, which distinguishes it from every earlier intellectual property treaty. Three provisions set the frame.

1. **Article 27.1** requires patents for inventions in all fields of technology if they are new, involve an inventive step and are capable of industrial application.
2. **Article 27.2** permits exclusion where preventing commercial exploitation is necessary to protect **public order or morality**, including life, health and the environment. This is the parent of section 3(b).
3. **Article 27.3(b)** permits exclusion of **plants and animals other than micro organisms** and of **essentially biological processes** for their production, but requires Members to protect **plant varieties** by patents, an **effective sui generis system**, or a combination.

Article 27.3(b) does four things at once. It **permits** exclusion of plants and animals. It **requires** patentability of micro organisms. It **permits** exclusion of essentially biological processes but not of microbiological ones. And it **mandates** some plant variety protection while leaving the form open. **That is the exact legal basis of India's two track answer: no patents on plants, and a special plant variety statute instead.**

The provision also required its own review four years after TRIPS came into force, that is in 1999. **That review began in 1999 and has never been completed.** Under paragraph 19 of the Doha Ministerial Declaration of 2001 the TRIPS Council was also to examine the relationship between TRIPS and the Convention on Biological Diversity, and the protection of traditional knowledge and folklore, and those three items remain its standing agenda.

The developing country demand, led by **India and Brazil**, is **mandatory disclosure of origin** in three parts: the country of origin or source of the genetic resources and traditional knowledge used, evidence of **prior informed consent**, and evidence of **fair and equitable benefit sharing**. Developed countries resist, arguing that disclosure belongs to contract and national access law and that a patent sanction would create uncertainty of title. India's objection has two limbs: whether patenting life is ethically acceptable, and the need to recognise **informal systems of innovation**, meaning community and farmer innovation and not only laboratory work.

The answer came from another forum. The **WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge** was adopted at Geneva on **24 May 2024**, the first treaty of that organisation on this interface and the first with provisions specifically for indigenous peoples and local communities. Four articles matter.

1. **Article 3** creates a **mandatory patent disclosure requirement** where the claimed invention is based on genetic resources or associated traditional knowledge.
2. **Article 4** makes the obligation **non retroactive**.
3. **Article 5** requires **effective and proportionate sanctions**, but provides that a patent shall not be revoked or rendered unenforceable solely for failure to disclose, except in cases of fraudulent intent.
4. **Article 17** provides for entry into force three months after **fifteen ratifications**, not yet reached.

India already has the substance at home, in the disclosure of source and geographical origin required by section 10(4) of the Patents Act, with non disclosure and anticipation by traditional knowledge as grounds of opposition and revocation under sections 25 and 64. **India has been implementing unilaterally for twenty five years what it has been asking TRIPS to make universal.**

5.2 Indian exclusions from patentability

Section 2(1)(j) defines an invention as a new product or process involving an inventive step and capable of industrial application, and section 2(1)(ja) requires the inventive step to involve technical advance or economic significance **and** to be non obvious, which is stricter than in several other countries. Six exclusions matter here.

1. **Section 3(b).** Inventions contrary to **public order or morality**, or seriously prejudicial to human, animal or plant life or health or to the environment.
2. **Section 3(c).** Mere discovery of a scientific principle, an abstract theory, or **any living thing or non living substance occurring in nature**. India's product of nature rule, which catches an isolated natural gene sequence.
3. **Section 3(d).** A new form of a known substance that does not **enhance known efficacy**, with an Explanation deeming salts, esters, polymorphs, isomers and metabolites to be the same substance unless they differ significantly in properties with regard to efficacy.
4. **Section 3(e).** A mere **admixture**, which is why a fixed dose combination must show synergy and not mere addition.
5. **Section 3(i).** All **methods of treatment** of humans, and of animals to free them from disease or increase economic value.
6. **Section 3(p).** What is in effect **traditional knowledge**.

Section 3(j) is the centre of this Part. It excludes *plants and animals in whole or any part thereof other than micro organisms, but including seeds, varieties and species, and essentially biological processes for production or propagation of plants and animals*. It contains four exclusions and one carve out.

1. The **whole plant or animal** is excluded.
2. **Any part thereof** is excluded, which is wider than most jurisdictions and arguably reaches a cell or a seed.
3. **Seeds, varieties and species** are expressly excluded, so the plant variety statute is the only route.
4. **Essentially biological processes**, meaning crossing and selection, are excluded.
5. The carve out is **micro organisms**, which India was obliged to allow and did allow by the 2002 amendment.

The open question is where a **constructed DNA sequence** falls. It is not a plant, an animal, a seed, a variety or a micro organism. Is it a laboratory chemical outside section 3(j), or does it become an inseparable part of a plant once inserted, so that a patent on it is in substance a patent on a plant part? **That is the question the Supreme Court declined to answer in Monsanto.**

5.3 Section 3(d): the leading case

Novartis AG v. Union of India, (2013) 6 SCC 1 (Supreme Court, 1 April 2013)

Facts. Novartis sought a patent on the **beta crystalline form of imatinib mesylate**, sold as Glivec for chronic myeloid leukaemia. The free base was already known. The claim was refused by the Controller, the refusal was upheld by the appellate board, and the matter reached the Supreme Court.

Held. The appeal was dismissed. In the case of a medicine, **efficacy in section 3(d) means therapeutic efficacy**, strictly judged. Better flow, stability, lower hygroscopicity, and even improved bioavailability standing alone, do not establish it without evidence of therapeutic benefit.

Principle. Section 3(d) is a **deliberate second tier of scrutiny beyond novelty and inventive step, aimed at evergreening**. It does **not** make incremental innovation unpatentable, and bioavailability is not always irrelevant; the applicant must simply prove enhanced therapeutic efficacy on evidence. A separate constitutional challenge to section 3(d) was dismissed by the Madras High Court in August 2007 and not appealed.

5.4 Patenting life: the global cases

Diamond v. Chakrabarty, 447 U.S. 303 (1980) (Supreme Court of the United States)

Facts. A genetically engineered *Pseudomonas* bacterium could break down several components of crude oil, useful for treating oil spills. The Patent Office allowed the process claims but refused the claim to the bacterium itself, holding that living things are not patentable.

Held. By five to four the Court held that a **live, human made micro organism is patentable subject matter** as a manufacture or composition of matter under the patent statute. The plant patent statutes did not show any intention to exclude living things, and hazard arguments must go to Congress.

Principle. **The line is not between living and inanimate things, but between products of nature and human made inventions.** This decision opened the biotechnology industry, and India reaches the same position on micro organisms through TRIPS and the 2002 amendment.

Association for Molecular Pathology v. Myriad Genetics, 569 U.S. 576 (2013) (Supreme Court of the United States)

Facts. Myriad located and sequenced the **BRCA1 and BRCA2** genes, mutations in which sharply raise the risk of breast and ovarian cancer, and patented the isolated genes, giving it control of diagnostic testing.

Held. A unanimous Court held that a **naturally occurring DNA segment is a product of nature and not patent eligible merely because it has been isolated**, but that **complementary DNA is patent eligible**, because the technician creates something that does not occur in nature.

Principle. The line is between **discovery and creation**. India reaches the same result on isolated natural DNA by the statutory route of **section 3(c)** rather than by judicial reasoning.

Harvard College v. Canada (Commissioner of Patents), 2002 SCC 76 (Supreme Court of Canada)

Facts. Harvard sought a patent on the **oncomouse**, a transgenic mouse bred to be susceptible to cancer and therefore useful in research, claiming both the process and the animal itself. The Commissioner allowed the process and refused the animal.

Held. By five to four the Court held that a **higher life form is not a manufacture or composition of matter** and so is not an invention. Any extension of that kind is for Parliament, given the public policy involved. The Commissioner has **no independent public policy discretion** to refuse a patent, so he reasoned wrongly while reaching the right result. A patent on amended claims issued in October 2003.

Principle. **The patentability of higher life forms is a legislative choice, not a judicial one.** Contrast *Chakrabarty*: the same statutory words produced opposite answers because the subject was a bacterium in one case and a mammal in the other.

The same mouse produced a third technique in Europe. In decision **T 19/90** of 3 October 1990 the Technical Board of Appeal of the European Patent Office set aside the refusal and remitted the application, holding that the exclusion of animal *varieties* is not a general ban on patenting animals and that the morality clause requires a **balance of animal suffering and environmental risk against usefulness to mankind**. The patent was granted in 1992, restricted to rodents on opposition, and further restricted to mice by decision **T 315/03** of 6 July 2004 before lapsing.

So four jurisdictions answered one question by four techniques. The United States asked whether the thing was made by man, Canada construed the statute and deferred to Parliament, Europe applied a morality and utility balance, and India legislated a categorical exclusion in section 3(j).

Monsanto Canada Inc. v. Schmeiser, 2004 SCC 34 (Supreme Court of Canada)

Facts. A farmer's fields contained Monsanto's patented herbicide resistant canola. He had not bought the seed or signed a technology agreement, and he saved and replanted the harvested seed. The patent claims were to the **gene and the modified cell**, not to the plant.

Held. By a narrow majority infringement was made out. By cultivating a crop containing the patented gene and cell the farmer **used** the invention and deprived the patentee of the full enjoyment of its monopoly. The *Harvard College* argument failed because the claims were not to the whole plant. On remedy the majority reversed the account of profits, since no profit was attributable to the invention.

Principle. **Claiming the gene rather than the plant lets a patent bite on a growing crop even where the plant itself is unpatentable.** This is exactly the strategy attempted in *Monsanto v. Nuziveedu*, and it succeeded in Canada while remaining undecided in India.

Bowman v. Monsanto Co., 569 U.S. 278 (2013) (Supreme Court of the United States)

Facts. A farmer bought commodity soybeans from a grain elevator, which were mostly patented herbicide resistant seed, planted them, sprayed with herbicide and saved seed from successive harvests for replanting.

Held. A unanimous Court held that **patent exhaustion does not permit a farmer to reproduce patented seed by planting and harvesting**, because exhaustion protects the particular article sold and does not authorise making new copies. The Court warned that the holding may not fit every self reproducing technology.

Principle. Exhaustion protects **use and resale of the thing bought, not manufacture of new copies.**

Compare India: **section 39(1)(iv) of the plant variety Act expressly lets a farmer save, use, sow, resow, exchange, share and even sell farm produce including seed of a protected variety, barring only branded seed.** India has legislated the opposite answer for registered varieties.

5.5 Dimminaco: the Indian breakthrough

Dimminaco A.G. v. Controller of Patents and Designs, Calcutta High Court, 15 January 2002

Facts. A Swiss company sought a patent for a **process for preparing an infectious bursitis vaccine** for poultry. The end product contained a **live attenuated virus**. The Assistant Controller refused the application on 27 December 1999, holding that a process yielding a living end product is not a manufacture, that a vaccine containing a living organism is not a substance, and that no such patent had ever been granted in India.

Held. The refusal was **quashed** and reconsideration directed within two months. **Manufacture is not defined** in the Act, so its ordinary or trade meaning applies where the end product is a commercial entity. There is **no statutory bar** to patenting a process merely because the end product contains a living organism. The Patent Office had in fact already granted such patents, and the argument that freeze drying kills the organism failed, because **lyophilisation preserves biological activity**.

Principle. The **vendibility test** governs: it is satisfied where the invention produces a vendible item, or improves, restores or preserves one. **This is the decision that made Indian biotechnology process patenting possible, and it came before the 2002 amendment expressly allowed micro organism patents.** Note two limits: it was a **process** claim, not a product claim to the organism, and the Court said the claim would still have to be tested under section 3.

5.6 Monsanto v. Nuziveedu: the central Indian case

Monsanto Technology LLC v. Nuziveedu Seeds Ltd., (2019) 3 SCC 381 (Supreme Court, 8 January 2019)

Facts. Monsanto held Indian **patent No. 214436**, with claims 1 to 24 for processes and claims 25 to 27 for a **nucleic acid sequence**. A sublicense of 21 February 2004 let Indian seed companies develop engineered hybrid cotton seed against a trait value. After State price control, the licensees paid the statutory rate instead of the contractual rate, and Monsanto terminated on 14 November 2015 and sued for patent and trade mark infringement. The defendants counterclaimed for revocation under **section 3(j)**. Monsanto argued the sequence was a man made laboratory chemical that could be removed again; the defendants argued it becomes an inseparable and inheritable part of the plant.

Held. The Supreme Court **set aside the Division Bench, restored the Single Judge's order of 28 March 2017 and remanded the suit**. The Division Bench should have confined itself to whether the injunction was justified and should not have examined the counterclaim, because **summary adjudication of a technically complex suit requiring expert evidence, at the injunction stage, is neither desirable nor permissible**. Revocation under **section 64** presupposes proper consideration on evidence and not decision from text books. The Court expressly **left all questions of fact and law open**, including whether the inserted sequence can be removed from the variety and whether the patented sequence is a plant or a part of a plant.

Principle. **Patentability and the section 3(j) exclusion are mixed questions of law and fact needing expert evidence and a trial.** Do not say the Court upheld the patent, or held gene sequences patentable, or held them excluded, or upheld the Division Bench; all four are wrong, and the last is a common error in secondary sources. The criticism is that by deciding on procedure the Court passed over the substance

and never engaged with Article 27.3(b), so India still has no authoritative answer on gene construct patents.

One extra point earns marks. The case shows how price regulation, contract, patent law and plant variety law interact, because Monsanto's exposure arose not from any defect in its patent but from a **price control order** that made its contractual trait value unenforceable, a position its own senior counsel conceded.

5.7 Protection of Plant Varieties and Farmers' Rights Act, 2001

India was obliged to protect plant varieties, chose neither patents nor membership of the international union, and used the words **effective sui generis system** to build in **farmers' rights**, which no version of the UPOV Convention contains as a positive right. The Act was passed in 2001 and registration became operational in 2006 and 2007.

Four kinds of variety can be registered: **new**, **extant**, **farmers'** and **essentially derived**. The criteria are novelty for a new variety plus **distinctiveness, uniformity and stability**, tested by growing the candidate beside the most similar known varieties over at least two seasons. **There is no requirement of inventive step or industrial application, and that is the central difference from a patent.**

Section 18 encodes the whole policy of the Act in five requirements for every application.

1. **Clause (c).** An affidavit that the variety contains no gene or gene sequence involving **terminator technology**.
2. **Clause (e). Complete passport data of the parental lines**, with the geographical location in India from which the material was taken and the contribution of any farmer, village community, institution or organisation.
3. **Clause (f).** A statement of novelty, distinctiveness, uniformity and stability.
4. **Clause (g).** The prescribed fees.
5. **Clause (h).** A declaration that the genetic material was **lawfully acquired**.

Clauses (c), (e) and (h) work as an access, benefit sharing and anti biopiracy filter placed inside an intellectual property statute, and that is the most distinctive feature of Indian plant variety law.

The sections to be able to name are these.

1. **Section 15.** Registrable varieties and the criteria.
2. **Sections 20 and 21.** Acceptance, publication and opposition.
3. **Section 24.** Certificate and duration: eighteen years for trees and vines, fifteen years for other and extant varieties.
4. **Section 26. Benefit sharing** determined by the Authority after inviting claims, payable into the National Gene Fund.
5. **Section 28.** The breeder's exclusive rights to produce, sell, market, distribute, import and export.
6. **Section 30. Researchers' rights** to use a registered variety for experiment or research and as an initial source for new varieties.
7. **Section 34.** Grounds of **revocation**: incorrect information, ineligibility of the grantee, failure to furnish information, and grant not in the public interest.
8. **Section 39. Farmers' rights**, including registration of a farmers' variety, reward from the Gene Fund, and the seed freedom described above, with a duty under section 39(2) to disclose expected performance and compensate for failure.
9. **Section 41.** Community claims on behalf of a village or local community.
10. **Section 44. Exemption of farmers from fees** in all proceedings.
11. **Section 45.** The **National Gene Fund**.
12. **Section 47. Compulsory licence** three years after registration where public requirements for seed are unmet or seed is not available at a reasonable price.

PepsiCo India Holdings v. Kavitha Kuruganti, 2023 SCC OnLine Del 3912 and 2024 SCC OnLine Del 153 (Delhi High Court)

Facts. PepsiCo registered the chipping potato variety **FL 2027**, used for Lay's crisps. In its 2011 application it mistakenly ticked **new variety** instead of **extant variety** and gave first commercialisation as 2009 in India, when the variety had first been sold in Chile in 2002. The Registrar treated it as extant and registered it in 2016. After PepsiCo sued Gujarat potato farmers in 2018 and 2019 and withdrew the suits in May 2019, a farmers' rights campaigner applied under **section 34**.

Held. Three stages. In **December 2021** the Authority **revoked** the registration for incorrect information, ineligibility, failure to furnish documents including an unstamped assignment deed, and public interest. On **5 July 2023** a Single Judge **upheld** the revocation while rejecting the public interest ground. On **9 January 2024** a Division Bench **set that aside in substantial part**, treated the categorisation error as remediable, held that suing farmers is not by itself against public interest, and restored the renewal application to the Registrar.

Principle. **Section 34 has real teeth, and a farmers' organisation has standing to invoke it**, but the appellate court drew a line between a **procedural error** and **disentitling misconduct**. This is the first substantial revocation litigation under the Act. *Reports differ on the outcome, so always set out all three stages.*

On the international union, **UPOV 1978** allowed a limited farmer's privilege by silence, while **UPOV 1991** strengthened the breeder's right, extended it to harvested material and essentially derived varieties, and reduced the farmer's privilege to an optional exception that does **not** permit sale. India has not joined, and could not join the 1991 text without abandoning section 39(1)(iv).

5.8 Biological Diversity Act, 2002 and benefit sharing

The Act implements the **Convention on Biological Diversity, 1992**, whose three objectives are conservation, sustainable use and the **fair and equitable sharing of benefits** from genetic resources, supplemented by the **Nagoya Protocol of 2010**. Biological resources means plants, animals and micro organisms, their parts, genetic material and byproducts, but not human genetic material. Note the deliberate asymmetry in the scheme.

1. **Section 3.** A **foreign** person or entity needs **previous approval of the National Biodiversity Authority** before obtaining an Indian biological resource or associated knowledge for research, commercial use or bio survey.
2. **Section 7.** An **Indian** citizen or body corporate needs only **prior intimation to the State Biodiversity Board**, which may prohibit or restrict the activity.
3. **Section 4.** No transfer of research results abroad without approval.
4. **Section 6.** No application for an **intellectual property right** based on an Indian biological resource without approval, obtainable after acceptance but before sealing of a patent.
5. **Section 21.** The Authority determines **equitable benefit sharing** on mutually agreed terms, which may include joint ownership of rights, technology transfer, location of research units, association of Indian scientists and local people, venture funds or money.
6. **Section 40.** Power to exempt **normally traded commodities**, which generates most practical disputes.
7. **Section 41.** **Biodiversity Management Committees** with People's Biodiversity Registers.

Divya Pharmacy v. Union of India, 2018 SCC OnLine Utt 1035 (Uttarakhand High Court, 21 December 2018)

Facts. Divya Pharmacy, the manufacturing arm of a trust making Ayurvedic and nutraceutical products, used local biological resources as its main raw material. The Uttarakhand State Biodiversity Board demanded fair and equitable benefit sharing. The company argued that as a wholly Indian entity with no foreign participation it owed nothing, since the obligation applied only to foreign entities.

Held. The petition was **dismissed**. **Biological resources are the property of the nation**. A distinction between purely Indian and non Indian commercial use would be absurd if indigenous communities are to

be recognised as stewards of biodiversity, and State Boards exercise **regulatory** and not merely advisory power under the Act and the 2014 Guidelines.

Principle. Fair and equitable benefit sharing applies to Indian entities as much as to foreign entities. The decision has since been effectively codified, because section 7 as amended read with the 2025 Regulations expressly requires Indian entities to pay.

The **Biological Diversity (Amendment) Act, 2023**, effective April 2024, made five changes. It **decriminalised** most violations, replacing imprisonment with monetary penalties. It recognised **codified traditional knowledge** and exempted registered AYUSH practitioners and users of such knowledge from benefit sharing in specified contexts. It **simplified compliance** for start ups, small enterprises and AYUSH manufacturers. It **exempted users of notified cultivated medicinal plants** from benefit sharing, to shift pressure away from wild collection. And it streamlined the intellectual property route, so a foreign entity needs prior approval while an Indian entity need only register but must obtain approval before **commercialising** the right.

The criticisms are that the exemptions, especially for cultivated medicinal plants, may be used to avoid sharing with the communities that conserved those plants, that the line between codified and non codified knowledge is unworkable, and that monetary penalties weaken deterrence where local monitoring is already thin.

The **Access and Benefit Sharing Regulations, 2025**, notified on **29 April 2025**, replaced the 2014 Guidelines. Their key change is that **digital sequence information is expressly brought within scope**, closing the loophole by which a company could sequence a resource once and then work from the data forever. They also introduce turnover based slabs with an exemption at the lower end, retain the cultivated medicinal plants exemption, and allow conditional approval for intellectual property applications.

5.9 Biopiracy: three examples and the lesson

These three are **patent office proceedings, not court judgments**, and an answer that calls them cases is imprecise.

1. **Turmeric.** United States patent 5,401,504, granted in March 1995 for the use of turmeric in wound healing, was challenged by India's Council of Scientific and Industrial Research in October 1996 with thirty two prior art references in Sanskrit, Urdu and Hindi. **All six claims were cancelled by order of 20 November 1997.**
2. **Neem.** European patent EP 0436257, granted in September 1994 for controlling fungi with extracted neem oil, was opposed by a member of the European Parliament, an Indian research foundation and an international organic agriculture federation. It was **revoked in May 2000** for lack of novelty and inventive step in view of long prior use in India, and the revocation was **upheld on appeal in March 2005.**
3. **Basmati.** United States patent 5,663,484, *Basmati rice lines and grains*, granted to RiceTec on 2 September 1997, was challenged by India's export development authority in June 2000. A reexamination certificate of **29 January 2002** cancelled the broad claims covering the rice plant and **changed the title so that it no longer referred to basmati.**

The lesson is why these three are in the syllabus at all. **All three were defensive victories won at great expense after the patent had already been granted.** India therefore built three preventive mechanisms: the **Traditional Knowledge Digital Library**, which puts documented Indian knowledge before foreign patent offices; the **disclosure of source and geographical origin** requirement in the Patents Act, with opposition and revocation attached; and the **access and benefit sharing** regime under the biodiversity statute. The 2024 WIPO Treaty tries to make the second of these universal.

5.10 Digital sequence information

Digital sequence information is sequence data derived from a genetic resource and held in digital form. There is still **no globally agreed definition.**

It breaks the old model. Access and benefit sharing was built around **physical access**: someone comes to a country, takes a sample, and a contract regulates the taking. Once a sequence is in a public database,

anyone can use it for research, product development or computational design without touching the physical resource, and so without triggering any obligation. Two values then collide. **Open access** to sequence data is a real public good that drove the COVID vaccine response. **Benefit sharing** is a matter of equity for countries rich in biodiversity and poor in technology. **Any solution must keep access open while separating benefit sharing from access.**

The international answer came in three steps.

1. **December 2022.** At the fifteenth Conference of the Parties the Members decided to create a **multilateral mechanism** with a global fund, recognising open access as a key principle.
2. **November 2024.** At the sixteenth Conference at Cali they adopted the modalities and established the **Cali Fund**, covering sequence data that is publicly available, not already covered by a bilateral agreement and not covered by another treaty, and naming the biopharmaceutical industry among the benefiting sectors.
3. **25 February 2025.** The Fund was **launched** so that contributions could begin.

Contributions are set at **indicative rates of one per cent of annual profits or 0.1 per cent of annual revenue**, they are **voluntary and not legally mandated**, the methodology is still unsettled, and half the Fund is to go to indigenous peoples and local communities.

India legislated ahead of the consensus by covering sequence information in the 2025 Regulations. The patent question remains open: under the 2024 WIPO Treaty disclosure is required where the invention is **based on** a genetic resource, and nobody yet knows how far that phrase reaches, from a sequence, to a synthesised molecule, to a molecule designed by a model trained on millions of sequences. **That last case is the real frontier and no instrument answers it.** The ready made critique is that the mechanism is voluntary, the rates indicative, the definition undecided and the scope full of exclusions, so it is a framework awaiting content; yet it is the first instrument anywhere to accept that **information about a resource, and not only the resource itself, can carry a benefit sharing obligation.**

PART 6 CASE LAW BY TOPIC

6.1 Note on verification

Every citation below was checked against the judgment text, the official reporter, the court's own record, or the official database of the tribunal or patent office concerned. Bench and date are given so that each entry can be traced independently. Three cautions apply and are flagged where they arise: the *Dimminaco* judgment copy carries a typographical date, secondary reports conflict on the operative outcome of the *PepsiCo* appeal, and the bench in *Paschim Banga Khet Mazdoor Samity* is taken from standard reporters rather than the judgment itself.

6.2 Research ethics and human participants

1. ***United States v. Karl Brandt and others*** (Doctors' Trial), Nuremberg Military Tribunal I, Case No. 1; indictment 25 October 1946; **judgment 20 August 1947**; presiding judge Walter B. Beals
Sixteen convicted, seven acquitted, seven executed. The section headed *Permissible Medical Experiments* is the **ten point Nuremberg Code**, the first international code of research ethics.
2. ***Kalpana Mehta v. Union of India*, (2018) 7 SCC 1**; Supreme Court, Constitution Bench of five (Dipak Misra CJI, Khanwilkar, Sikri, Chandrachud and Ashok Bhushan JJ); **9 May 2018**
A Parliamentary Standing Committee report **may be relied upon** under Articles 32, 136 and 226, but **its correctness cannot be challenged**, because of privilege under Articles 105, 121 and 122. Arose from the human papillomavirus vaccine study in Khammam and Vadodara.

6.3 Animal research and animal welfare

1. ***Animal Welfare Board of India v. A. Nagaraja*, (2014) 7 SCC 547**; Supreme Court (Radhakrishnan and Pinaki Chandra Ghose JJ); **7 May 2014**

Jallikattu and bullock cart races as then conducted violated sections 3 and 11 of the 1960 Act read with Articles 51A(g), 48A and 21. Adopted the **five freedoms** and held that animals have a **right to live with dignity**.

2. **Animal Welfare Board of India v. Union of India**, 2023 SCC OnLine SC 661; Supreme Court, Constitution Bench of five, judgment by **Aniruddha Bose J**; **18 May 2023**
Upheld the Tamil Nadu, Maharashtra and Karnataka amendments as relatable to **Entry 17 of the Concurrent List** and as substantially reducing pain. **Nagaraja** was **confined, not overruled**, and the Article 29(1) cultural rights question was left open.

6.4 Environmental and constitutional principles

1. **Vellore Citizens Welfare Forum v. Union of India**, (1996) 5 SCC 647; AIR 1996 SC 2715; Supreme Court (Kuldip Singh, Faizan Uddin and Venkataswami JJ); **28 August 1996**
The **precautionary principle** and the **polluter pays principle** are part of the law of the land and of customary international law, with the **burden on the polluter** to show that its activity is harmless.
2. **Paschim Banga Khet Mazdoor Samity v. State of West Bengal**, (1996) 4 SCC 37; AIR 1996 SC 2426; Supreme Court (S. C. Agrawal and G. B. Pattanaik JJ); **6 May 1996**
Article 21 obliges the State to preserve life, so a government hospital and its doctors must give timely emergency treatment. Compensation was awarded for refusal. *Confirm the bench from SCC if you cite it.*

6.5 Genetically modified crops and biosafety

1. **Gene Campaign v. Union of India**, 2024 INSC 545; Supreme Court (B. V. Nagarathna and Sanjay Karol JJ); **23 July 2024**
Split verdict on genetically modified mustard DMH 11. Both judges agreed that **judicial review of decisions on genetically modified organisms is permissible**, and directed a **national policy** after a **national consultation within four months**, verification of experts' credentials, mitigation of conflicts of interest, and enforcement of the Food Safety and Standards Act on imports. Referred to the Chief Justice for a larger Bench.
2. **Confédération paysanne v. Premier ministre**, Court of Justice of the European Union, Grand Chamber; ECLI:EU:C:2018:583; **25 July 2018**
Organisms obtained by **mutagenesis are genetically modified organisms** under Directive 2001/18, the exemption covering only techniques conventionally used with a long safety record. Gene editing therefore falls inside the full European regime.
3. **European Communities, Measures Affecting the Approval and Marketing of Biotech Products**, WT/DS291/R, WT/DS292/R and WT/DS293/R; panel chaired by Christian Häberli; reports 29 September 2006, **adopted 21 November 2006**; not appealed
A **de facto moratorium from June 1999 to August 2003** and delay in **twenty four of twenty seven** products caused **undue delay** under Annex C(1)(a) and Article 8 of the SPS Agreement; nine national safeguard measures breached Articles 5.1 and 2.2 for want of a risk assessment. A Member may regulate strictly but may not refuse to decide.

6.6 Patentability of life forms: India

1. **Dimminaco A.G. v. Controller of Patents and Designs**, High Court at Calcutta, AID No. 1 of 2001 (Asok Kumar Ganguly J); **15 January 2002**, commonly reported as 2002 IPLR 255 (Cal)
Manufacture is undefined, so its ordinary meaning applies, and there is **no statutory bar** to a process patent merely because the end product contains a living organism. The **vendibility test** governs. *The judgment copy shows 15.01.2001, an obvious typographical error repeated by many sources.*
2. **Monsanto Technology LLC v. Nuziveedu Seeds Ltd.**, (2019) 3 SCC 381; AIR 2019 SC 559; Supreme Court (Nariman and Navin Sinha JJ, judgment by Navin Sinha J); **8 January 2019**
Section 3(j) and patentability are mixed questions of law and fact requiring expert evidence, so summary adjudication at the injunction stage was impermissible and revocation under section 64 presupposes consideration on evidence. Division Bench set aside, Single Judge's order restored, suit remanded. Patent No. 214436.

3. **Novartis AG v. Union of India**, (2013) 6 SCC 1; AIR 2013 SC 1311; Supreme Court (Aftab Alam and Ranjana Prakash Desai JJ); **1 April 2013**
Efficacy in section 3(d) means therapeutic efficacy, strictly judged; improved bioavailability alone is insufficient. Section 3(d) is a deliberate second tier of scrutiny against **evergreening**.

6.7 Patentability of life forms: foreign courts and the European Patent Office

1. **Diamond v. Chakrabarty**, 447 U.S. 303 (1980); Supreme Court of the United States, Burger CJ for a majority of five; **16 June 1980**
A live, human made micro organism is patentable. The line is **products of nature against human made inventions**, and hazard arguments belong to the legislature.
2. **Association for Molecular Pathology v. Myriad Genetics**, 569 U.S. 576 (2013); Supreme Court of the United States, Thomas J for a unanimous Court; **13 June 2013**
Isolated naturally occurring DNA is not patent eligible merely because isolated; **complementary DNA is**, because it does not occur in nature. Concerned the BRCA1 and BRCA2 patents.
3. **Harvard College v. Canada (Commissioner of Patents)**, 2002 SCC 76, [2002] 4 SCR 45; Supreme Court of Canada, Bastarache J for a majority of five, Binnie J dissenting; **5 December 2002**
A higher life form is not a manufacture or composition of matter and so is not an invention. Extension is for Parliament, and the Commissioner has **no independent public policy discretion**.
4. **Monsanto Canada Inc. v. Schmeiser**, 2004 SCC 34, [2004] 1 SCR 902; Supreme Court of Canada, McLachlin CJ and Fish J for a majority of five; **21 May 2004**
Cultivating a crop containing the **patented gene and cell** is **use** of the invention and infringes, consistently with **Harvard College** because the claims were not to the whole plant. Account of profits reversed for want of attributable profit.
5. **Bowman v. Monsanto Co.**, 569 U.S. 278 (2013); Supreme Court of the United States, Kagan J for a unanimous Court; **13 May 2013**
Patent exhaustion does not permit a farmer to reproduce patented seed by planting and harvesting, because exhaustion protects the article sold and not the making of new copies.
6. **Harvard Oncomouse**, European Patent Office, Technical Board of Appeal, **T 19/90 of 3 October 1990** and **T 315/03 of 6 July 2004**
The exclusion of animal *varieties* is not a general ban on patenting animals, and the morality clause requires a **balance of animal suffering and environmental risk against usefulness to mankind**. The patent was granted in 1992, restricted to rodents, then to mice, and finally lapsed.

6.8 Seeds, farmers' rights and plant varieties

1. **PepsiCo India Holdings v. Kavitha Kuruganti**, PPV and FR Authority revocation **December 2021**; Delhi High Court Single Judge (Navin Chawla J) **2023 SCC OnLine Del 3912, 5 July 2023**; Division Bench (Yashwant Varma and Dharmesh Sharma JJ) **2024 SCC OnLine Del 153, 9 January 2024**
Revocation of potato variety **FL 2027** under **section 34** was upheld by the Single Judge and then set aside in substantial part by the Division Bench, which treated the categorisation error as remediable, held that suing farmers is not by itself against public interest, and restored the renewal application. *Reports conflict on the outcome, so set out all three stages.*

6.9 Drug regulation, patent linkage and access to medicines

1. **Bayer Corporation v. Union of India** (patent linkage), Delhi High Court Single Judge (Ravindra Bhat J) 18 August 2009; Division Bench **2010 (43) PTC 12 (Del)**, **9 February 2010**; Supreme Court dismissed the appeal in **December 2010**
The Drugs and Cosmetics Act and the Patents Act are **separate codes with different objects**, so **patent linkage is not part of Indian law** and the drug regulator is not a patent enforcer.
2. **Bayer Corporation v. Union of India** (compulsory licence, Nexavar), Controller 9 March 2012; appellate board 4 March 2013; Bombay High Court **2014 (60) PTC 277 (Bom)**, **15 July 2014**; Supreme Court declined to interfere on **12 December 2014**

India's **first compulsory licence** under **section 84**, satisfied on all three grounds. **Worked in the territory of India** is a fact dependent question and does not always require local manufacture.

6.10 Biodiversity, access and benefit sharing

1. **Divya Pharmacy v. Union of India, 2018 SCC OnLine Utt 1035**; Uttarakhand High Court (Sudhanshu Dhulia J); Writ Petition (M/S) No. 3437 of 2016; **21 December 2018**

Fair and equitable benefit sharing applies to Indian as well as foreign entities, biological resources are the property of the nation, and State Biodiversity Boards exercise regulatory power under the Act and the 2014 Guidelines.

6.11 Biopiracy: patent office proceedings, not cases

1. **Turmeric**, United States patent **5,401,504**, granted 28 March 1995; reexamination sought by the Council of Scientific and Industrial Research on 28 October 1996
All six claims cancelled by order of 20 November 1997 on prior art showing centuries of documented Indian use.
2. **Neem**, European patent **EP 0436257 B1**, granted 14 September 1994; revoked by the Opposition Division in May 2000; revocation upheld in **T 0416/01 of 8 March 2005**
Revoked for want of novelty and inventive step, in view of long prior public use of neem's antifungal properties in India.
3. **Basmati**, United States patent **5,663,484**, granted 2 September 1997; reexamination sought in June 2000; **certificate of 29 January 2002**
The **broad claims covering the rice plant were cancelled** and the title was changed so that it no longer referred to basmati. *Sources differ on whether the patent had 20 or 24 claims, so do not commit to a number.*

6.12 Authorities cited inside the leading cases

1. **Alka Gupta v. Narender Kumar Gupta, (2010) 10 SCC 141**
Relied on in **Monsanto: there are no short cuts in the trial of suits**, and a suit cannot be short circuited by deciding issues of fact on pleadings and documents without trial.
2. **Bishwanath Prasad Radhey Shyam v. Hindustan Metal Industries, (1979) 2 SCC 511; AIR 1982 SC 1444**
Relied on in **Dimminaco**: a patent requires **novelty and utility**, and the object of patent law is to **encourage scientific research, new technology and industrial progress**.

PART 7 QUICK REVISION

7.1 Dates that carry marks

1940 Drugs and Cosmetics Act. 1947 Nuremberg Code. 1960 Prevention of Cruelty to Animals Act. 1964 Declaration of Helsinki, last revised **19 October 2024**. 1970 Patents Act. 1979 Belmont Report. 1982 CIOMS guidelines. 1986 Environment (Protection) Act and the United States Coordinated Framework. **1989 Rules** for engineered organisms. 1992 Convention on Biological Diversity. 1994 TRIPS. 1998 animal experimentation Rules, amended 2001 and January 2006. **Cartagena Protocol adopted 29 January 2000, in force 11 September 2003**. 2001 Plant Varieties Act and the Doha Declaration. 2002 Biological Diversity Act, the Patents Amendment permitting micro organisms, **Dimminaco**, and Bt cotton approval. 2005 product patents for medicines. 2006 Food Safety and Standards Act and the World Trade Organization biotech reports. **9 February 2010 Bt brinjal moratorium**. 2012 Technical Expert Committee and India's first compulsory licence. 2013 **Novartis**. 2017 Medical Devices Rules and the ICMR ethical guidelines. **19 March 2019 New Drugs and Clinical Trials Rules**. February 2020 all devices notified as drugs. October 2022 engineered mustard cleared by the appraisal committee. 11 August 2023 Jan Vishwas Act. **24 May 2024 WIPO Treaty; 23 July 2024 Gene Campaign; 1 November 2024 the Cali Fund decision**. 29 April 2025 Access and Benefit Sharing Regulations.

7.2 Ten mistakes that lose marks

1. Saying the Supreme Court in **Monsanto** decided whether gene sequences are patentable. **It refused to decide.**
2. Saying **Gene Campaign** banned or approved engineered mustard. **It split.**
3. Saying **Kalpana Mehta** held the vaccine trials illegal. **It decided a question about parliamentary reports.**
4. Giving the Belmont Report four principles. **It has three**, each with a matching application.
5. Saying Helsinki was last amended in 2000. **It was revised in October 2024.**
6. Describing the Drugs, Medical Devices and Cosmetics Bill as law. **It is a draft.**
7. Confusing **section 3(b)** on morality with **section 3(d)** on new forms and **section 3(j)** on plants and animals.
8. Confusing the **compulsory licence** on Nexavar with the **patent linkage** petition. Same drug, different questions.
9. Calling the turmeric, neem and basmati matters cases. **They were patent office proceedings.**
10. Saying terminator technology is banned outright by the plant variety Act. **The Act denies registration**, under sections 18(1)(c) and 29(3).